MANIPAL INSTITUTE OF TECHNOLOGY

Manipal Academy of Higher Education, Manipal

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

THIRD SEMESTER B.TECH. END SEMESTER EXAMINATION - 2025

CS 2103: DATA ANALYTICS

Branch: Computer Science & Engineering (CSE) | Credit: 4

Time: 3 Hours | Max. Marks: 50

Instructions: Answer ALL questions. Each question carries designated marks.

PART A

1. (a) Explain exploratory data analysis (EDA): outlier detection using IQR and box plots. [5 Marks]

1. (b) Formulate Ordinary Least Squares (OLS) Multiple Linear Regression and R-squared interpretation. [5 Marks]

PART B

2. (a) Explain logistic regression for binary classification and cross-entropy loss derivation. [5 Marks]

2. (b) Describe Principal Component Analysis (PCA) eigenvalue decomposition for dimensionality reduction. [5 Marks]

PART C

3. (a) Compare K-Means clustering algorithm and Hierarchical Agglomerative clustering with dendrograms. [10 Marks]

PART D

4. (a) Explain precision, recall, F1-score, ROC curves, and AUC in imbalanced datasets. [10 Marks]

[END OF QUESTION PAPER - MIT MANIPAL CSE 3RD SEMESTER ARCHIVE]